COSC2779/COSC2972 Deep Learning:
Exam Example Questions (No Solutions)

This document provides sample questions for the three question types that may appear in the COSC2779/COSC2972 Deep Learning Exam.

These examples are intended to help you understand the style and format of the exam questions. The actual exam will contain a different number of questions and different question content.

Students should use these examples as a guide only and ensure they are familiar with all course materials and learning outcomes assessed in the course.

PART A – MULTIPLE CHOICE QUESTIONS

The following multiple-choice questions may have multiple correct answers. You must choose ALL the correct options while NOT selecting any of the incorrect options. No explanation is required.

For each question, the penalty for selecting an incorrect option is equal to the mark awarded for a correctly selected option. The score for a question cannot be less than 0.

Question A.1:

Which of the following are properties of convolutional neural networks?

(i) Sparse connectivity (feature extraction usually happens locally)
(ii) Parameter sharing (the same operation is applied at different locations)
(iii) Dense connectivity (every output is connected to all input locations)
(iv) Pooling helps reduce redundant information and provides some level of invariance to translations

Question A.2:

During fine-tuning in transfer learning, which of the following are correct steps?

(i) Unfreeze some top layers of the base model
(ii) Train the unfrozen base model layers with a smaller learning rate
(iii) Both the head and base model weights can be trained
(iv) All layers in the base model must remain frozen

Question A.3:

You are given the task of predicting the genre of short music clips of different lengths. Which Recurrent Neural Network (RNN) architecture is most suitable for this task?

(i) One-to-one
(ii) One-to-many
(iii) Many-to-one
(iv) Many-to-many

Question A.4:

Which of the following are issues with a basic autoencoder?

(i) It can memorize (over-fit) the training data
(ii) It cannot reliably generate novel samples (e.g. images) because its latent space distribution is unconstrained
(iii) It cannot reconstruct its input
(iv) It requires adversarial training

PART B – SHORT ANSWER QUESTIONS

The questions in this section can be answered in 2–4 sentences. Please keep your responses concise and focused.

Question B.1:

You are designing a neural network for image classification. A colleague suggests replacing all convolutional layers with fully connected layers. Why might convolutional layers be more suitable for image data?

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Question B.2:

In a recurrent encoder–decoder model without attention, the decoder relies on the final encoder state to represent the entire input sequence. What limitation can this cause, particularly for long sequences, and how does attention help to address this limitation?

Question B.3:

You have a pre-trained Transformer and a small labeled dataset for a new classification task. You want to adapt the model while training as few parameters as possible. Would you use feature extraction or full fine-tuning? Briefly explain why and how you would do this.

PART C - PROBLEM-SOLVING QUESTIONS

The questions in this section involve multi-step problems based on practical deep learning scenarios. Answer each part clearly and show your reasoning where appropriate.

Problem C.1 - Developing a CNN Image Classifier

You are developing a deep learning system to classify images of recyclable waste into six categories. You have several thousand labelled RGB images and decide to develop a CNN-based classifier.

(a) Choosing a CNN

A team member suggests using a fully connected neural network directly on the image pixels instead of a CNN. Why is a CNN more suitable for this task? Give two reasons.

(b) Initial training

You split your dataset into training, validation, and test sets. For your first attempt, you train a CNN from scratch and obtain the following results:

- Training accuracy: 72%
- Validation accuracy: 70%

Both training and validation performance are lower than you expected. Does this result mainly suggest overfitting or underfitting? Give one change you could try to address this problem.

(c) Improving generalization

After increasing the capacity of the CNN and training it again, you obtain:

- Training accuracy: 96%
- Validation accuracy: 78%

What problem does this result suggest? Briefly explain why and give two techniques that could help address this problem.

(d) Evaluating the classification performance

After improving the model, you obtain good overall validation accuracy. However, you want to understand whether the CNN performs well across all six waste categories, and which categories it tends to confuse. What additional evaluation results could you examine, and what information would they provide?

(e) Using a pretrained model

You later obtain access to an ImageNet-pretrained ResNet. Instead of continuing to train your CNN from scratch, describe how you could adapt the pretrained ResNet to this classification task.